AVGO Broadcom Inc. — 个股深度研究 (专业中文版) v20261016-0756

> 数据截至 2026-10-16 07:56 SGT｜价格 $344.33｜市值 1,643.7 bn USD｜财报货币 USD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

博通是’XPU/TPU 替代 GPU’趋势的核心受益者，加上 Tomahawk/Jericho 以太网与 VMware 现金流；高客户集中度意味着单一客户项目变动影响巨大。

- **多头驱动**: Multiple XPU customers in volume; Ethernet scale-up/out wins; software stable｜多 XPU 客户放量；以太网胜出；软件稳定
- **空头驱动**: Customer in-sourcing (e.g., MediaTek on TPU), lumpy AI revenue, debt from VMware｜客户转单、AI 收入波动、VMware 债务
- **证伪条件 Falsifiers**: AI semi revenue guide decelerates; key XPU customer shifts vendor; VMware revenue declines｜AI 半导体指引放缓；主要 XPU 客户转供应商；VMware 收入下降

2. 产品、客户与收入模式

- 主题 Theme: Custom AI ASIC + networking + infrastructure software 定制 AI 芯片+网络+软件
- 业务分部 Segments: AI semiconductors (XPU+networking) AI 半导体；Non-AI semis 非 AI 半导体；Infrastructure software (VMware) 基础软件
- 公司描述 (DATA, Yahoo/公司申报): Broadcom Inc. designs, develops, and supplies various semiconductor devices and infrastructure software solutions internationally. The company operates in two segments: Semiconductor Solutions and Infrastructure Software. The company offers networking connectivity, such as custom silicon solutions, ethernet switching & routing, ethernet NIC controllers, physical layer devices, and fiber optic components; wireless device connectivity, including RF semiconductor devices, connectivity solutions, custom touch controllers, and inductive charging ASICS; servers and storage system solutions, such as PCIE switches, SAS & raid products, fibre channel products, and HDD & SSD solutions; broadband solutions, includes set-top box, and broadband access; and industrial. The company also offers a private cloud software

portfolio, including the VMware Cloud Foundation, Edge, vSphere foundation, telco clo

- 主要客户: Google (TPU), Meta, ByteDance, OpenAI, Anthropic, Apple; enterprises (VMware)｜谷歌 TPU、Meta、字节、OpenAI、Anthropic、苹果；企业(VMware)

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: TSMC (N3/N2, CoWoS), HBM vendors, substrate makers｜台积电、HBM、载板
- 竞争者 (含业外): NVDA, Marvell, Alchip/GUC/MediaTek (ASIC), Arista/Cisco (switching)｜NVDA、Marvell、世芯/创意/联发科、Arista/思科
- 瓶颈/监管: CoWoS allocation, customer concentration, VMware renewal cycle｜CoWoS 配额、客户集中度、VMware 续约
- 关键 KPI: AI revenue, XPU customers count, software revenue, FCF, net debt｜AI 收入、XPU 客户数、软件收入、FCF、净负债
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, USD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-07-31	29.6 bn USD	20.5 bn USD	16.1 bn USD	13.1 bn USD	13.7 bn USD	54.3%
2026-04-30	22.2 bn USD	15.4 bn USD	10.9 bn USD	9.3 bn USD	10.3 bn USD	48.9%
2026-01-31	19.3 bn USD	13.2 bn USD	8.7 bn USD	7.3 bn USD	8.0 bn USD	44.9%
2025-10-31	18.0 bn USD	12.2 bn USD	7.7 bn USD	8.5 bn USD	7.5 bn USD	42.5%
2025-07-31	16.0 bn USD	10.7 bn USD	6.1 bn USD	4.1 bn USD	7.0 bn USD	38.1%
2025-04-30	N/A	N/A	N/A	N/A	N/A	N/A
2025-01-31	N/A	N/A	N/A	N/A	N/A	N/A

资产负债: 现金 24.0 bn USD；总负债 59.4 bn USD；净现金 -35.4 bn USD。TTM FCF 39.4 bn USD（FCF 率 44.2%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-10-31	1.95	1.86817	0.0438
2026-01-31	2.05	2.02326	0.013200001
2026-04-30	2.44	2.39835	0.0174
2026-07-31	3.32	3.238	0.0253

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-12-10**；本季 EPS 一致预期 3.82818（40 位分析师）；收入一致预期 34.9 bn USD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$159.55	-53.7%
2Q	Base	$404.24	17.4%
2Q	Bull	$705.45	104.9%
4Q	Bear	$169.51	-50.8%
4Q	Base	$490.89	42.6%
4Q	Bull	$919.58	167.1%
6Q	Bear	$179.90	-47.8%
6Q	Base	$595.60	73.0%
6Q	Bull	$1,197.50	247.8%
8Q	Bear	$190.76	-44.6%
8Q	Base	$722.16	109.7%
8Q	Bull	$1,558.19	352.5%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	20.2x
4Q	15.7x
6Q	12.2x
8Q	9.5x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	32.1%	36.0%	13.6%	2.5%	25.0%	$179.74	-47.8%
Base	64.2%	45.0%	12.6%	3.0%	50.0%	$778.24	126.0%
Bull	83.4%	51.7%	12.1%	3.5%	25.0%	$1,837.47	433.6%

概率加权价值 **$893.42**（相对现价 159.5%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 3,750.5 bn USD；终值占比 57.1%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **39.9%**；一致预期 FY+1 增速 64.2%。
- WACC (CAPM) = rf 5.2% + β 1.48 × ERP 5.0% = 12.6%。
- SOTP segments 分部: AI semiconductors (XPU+networking) AI 半导体; Non-AI semis 非 AI 半导体; Infrastructure software (VMware) 基础软件. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | AVGO | 1,643.7 bn USD | 18.8x | 17.8x | 85.5% | 75.5% | 2.4% | | MRVL | 216.8 bn USD | 22.5x | 33.1x | 36.5% | 52.2% | 1.1% | | NVDA | 5,245.5 bn USD | 17.2x | 13.6x | 105.9% | 74.7% | 0.8% | | ANET | 234.1 bn USD | 20.9x | 35.5x | 37.7% | 63.0% | 1.7% | | QCOM | 190.5 bn USD | 4.4x | 17.5x | -4.0% | 54.2% | 5.4% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: Customer in-sourcing (e.g., MediaTek on TPU), lumpy AI revenue, debt from VMware｜客户转单、AI 收入波动、VMware 债务
- 证伪指标: AI semi revenue guide decelerates; key XPU customer shifts vendor; VMware revenue declines｜AI 半导体指引放缓；主要 XPU 客户转供应商；VMware 收入下降

- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	344.3	Yahoo Finance quote, fetched 2026-10-16T07:56 SGT
Market cap 市值	1.644e+12	Yahoo info.marketCap 2026-10-16T07:56
Cash 现金	2.398e+10	Yahoo info.totalCash (latest filing) 2026-10-16T07:56
Debt 负债	5.942e+10	Yahoo info.totalDebt (latest filing) 2026-10-16T07:56
Revenue TTM 收入	8.91e+10	Yahoo info.totalRevenue (TTM, USD) 2026-10-16T07:56
FCF TTM 自由现金流	3.94e+10	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-07-31', '2026-04-30', '2026-01-31', '2025-10-31'] (USD)
Beta	1.476	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.0519	FRED DGS10 2026-10-14
FX USD per USD	1	n/a (USD)
Consensus rev growth FY0	0.6588	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.6416	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market ERP
Beta clamp	[0.8,1.6] → 1.48	ASSUMPTION: limit extreme betas

Assumption	Value	Rationale
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%
Year-1 growth (Base)	0.6416	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION
Starting FCF margin	0.4422	DATA: FCF TTM / Revenue TTM, floored -50%
Long-run FCF margin (Base)	0.45	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

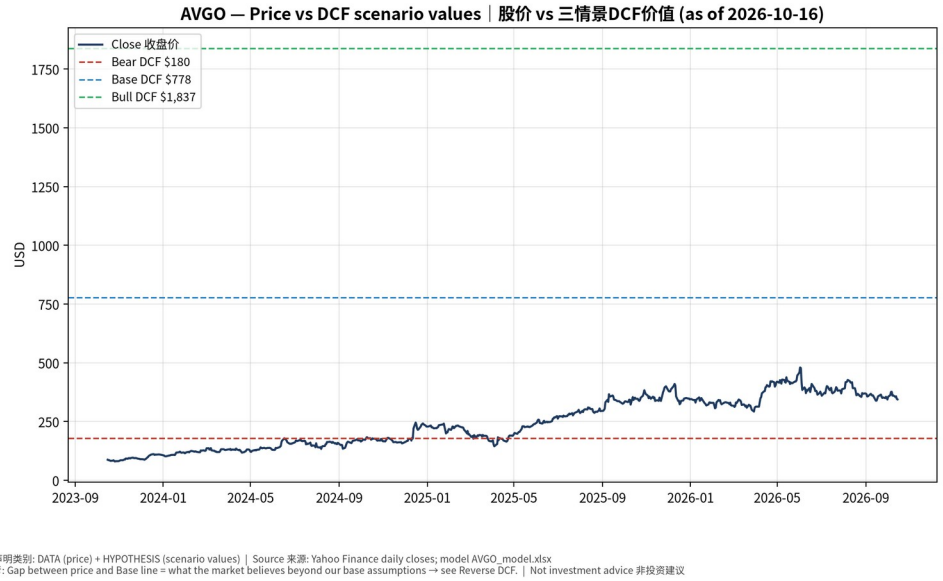
12. Model Provenance 模型溯源

- Model file: `AVGO_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT
- Version: v20261016-0756. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

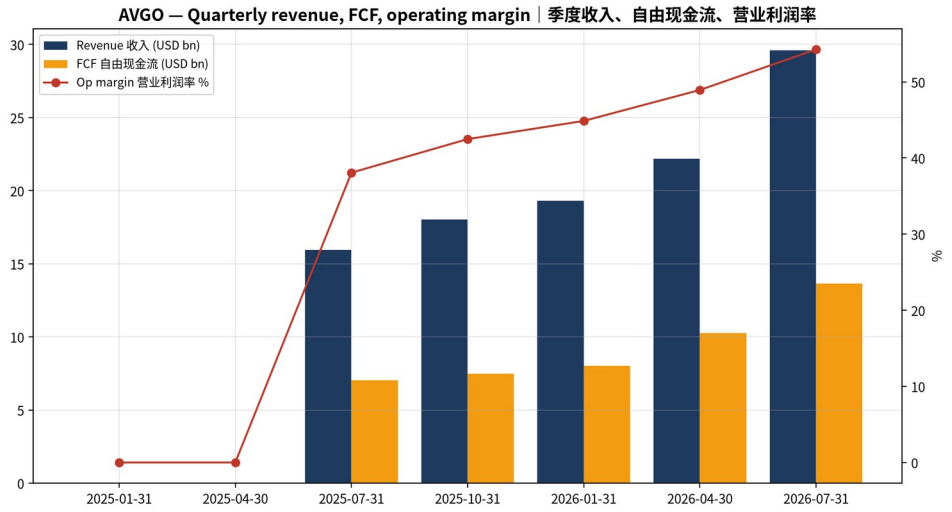
- v20261016-0756: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-12-10 (TENTATIVE)。

图片报告 Image Report



1_price_vs_value

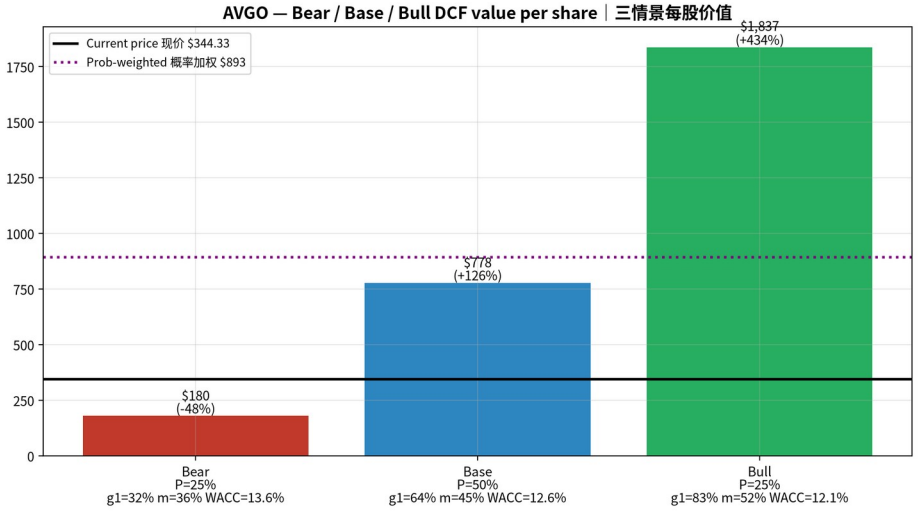
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

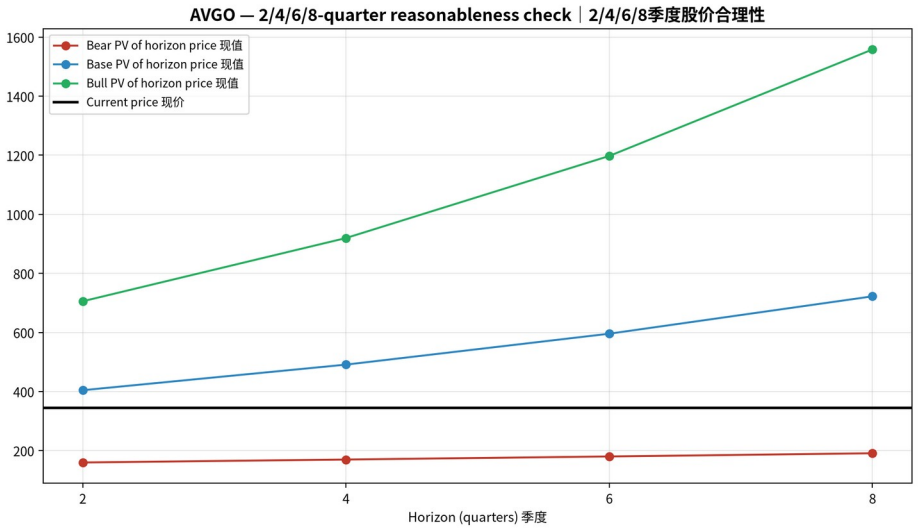
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验Horizon check



Reverse DCF 反向DCF: price implies Y1 growth ≈ 40%

Consensus FY+1 growth 一致预期: 64% → gap = expectation the price embeds 价格隐含预期差

Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

5_logic_map

研究逻辑图Logic map

AVGO Broadcom Inc. — Equity Deep Dive (Professional English) v20261016-0756

Data as of 2026-10-16 07:56 SGT | Price $344.33 | Mkt cap 1,643.7 bn USD | Reporting currency USD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

Broadcom is the core beneficiary of the XPU/TPU-vs-GPU substitution trend, plus Tomahawk/Jericho Ethernet and VMware cash flow; high customer concentration means single-program shifts matter a lot.

- **Bull drivers**: Multiple XPU customers in volume; Ethernet scale-up/out wins; software stable｜多 XPU 客户放量；以太网胜出；软件稳定
- **Bear drivers**: Customer in-sourcing (e.g., MediaTek on TPU), lumpy AI revenue, debt from VMware｜客户转单、AI 收入波动、VMware 债务
- **Falsifiers**: AI semi revenue guide decelerates; key XPU customer shifts vendor; VMware revenue declines｜AI 半导体指引放缓；主要 XPU 客户转供应商；VMware 收入下降

2. Products, customers, revenue model

- Theme: Custom AI ASIC + networking + infrastructure software 定制 AI 芯片+网络+软件 | Segments: AI semiconductors (XPU+networking) AI 半导体; Non-AI semis 非 AI 半导体; Infrastructure software (VMware) 基础软件
- Business description (DATA): Broadcom Inc. designs, develops, and supplies various semiconductor devices and infrastructure software solutions internationally. The company operates in two segments: Semiconductor Solutions and Infrastructure Software. The company offers networking connectivity, such as custom silicon solutions, ethernet switching & routing, ethernet NIC controllers, physical layer devices, and fiber optic components; wireless device connectivity, including RF semiconductor devices, connectivity solutions, custom touch controllers, and inductive charging ASICS; servers and storage system solutions, such as PCIE switches, SAS & raid products, fibre channel products, and HDD & SSD solutions; broadband solutions, includes set-top box, and broadband access; and industrial. The company also offers a private cloud software portfolio, including the VMware Cloud Foundation, Edge, vSphere foundation, telco clo
- Customers: Google (TPU), Meta, ByteDance, OpenAI, Anthropic, Apple; enterprises (VMware)｜谷歌 TPU、Meta、字节、OpenAI、Anthropic、苹果；企业(VMware)

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: TSMC (N3/N2, CoWoS), HBM vendors, substrate makers｜台积电、HBM、载板
- Competitors: NVDA, Marvell, Alchip/GUC/MediaTek (ASIC), Arista/Cisco (switching)｜NVDA、Marvell、世芯/创意/联发科、Arista/思科
- Bottlenecks/regulation: CoWoS allocation, customer concentration, VMware renewal cycle｜CoWoS 配额、客户集中度、VMware 续约
- KPIs: AI revenue, XPU customers count, software revenue, FCF, net debt｜AI 收入、XPU 客户数、软件收入、FCF、净负债
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, USD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-07-31	29.6 bn USD	20.5 bn USD	16.1 bn USD	13.1 bn USD	13.7 bn USD	54.3%
2026-04-30	22.2 bn USD	15.4 bn USD	10.9 bn USD	9.3 bn USD	10.3 bn USD	48.9%
2026-01-31	19.3 bn USD	13.2 bn USD	8.7 bn USD	7.3 bn USD	8.0 bn USD	44.9%
2025-10-31	18.0 bn USD	12.2 bn USD	7.7 bn USD	8.5 bn USD	7.5 bn USD	42.5%
2025-07-31	16.0 bn USD	10.7 bn USD	6.1 bn USD	4.1 bn USD	7.0 bn USD	38.1%
2025-04-30	N/A	N/A	N/A	N/A	N/A	N/A
2025-01-31	N/A	N/A	N/A	N/A	N/A	N/A

Cash 24.0 bn USD; debt 59.4 bn USD; TTM FCF 39.4 bn USD (FCF margin 44.2%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-10-31	1.95	1.86817	0.0438

Quarter	EPS actual	EPS consensus	Surprise
2026-01-31	2.05	2.02326	0.013200001
2026-04-30	2.44	2.39835	0.0174
2026-07-31	3.32	3.238	0.0253

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-12-10**; current-quarter EPS consensus 3.82818 (40 analysts); revenue consensus 34.9 bn USD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$159.55	-53.7%
2Q	Base	$404.24	17.4%
2Q	Bull	$705.45	104.9%
4Q	Bear	$169.51	-50.8%
4Q	Base	$490.89	42.6%
4Q	Bull	$919.58	167.1%
6Q	Bear	$179.90	-47.8%
6Q	Base	$595.60	73.0%
6Q	Bull	$1,197.50	247.8%
8Q	Bear	$190.76	-44.6%
8Q	Base	$722.16	109.7%
8Q	Bull	$1,558.19	352.5%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	20.2x
4Q	15.7x
6Q	12.2x
8Q	9.5x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bear	32.1%	36.0%	13.6%	2.5%	25.0%	$179.74	-47.8%

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Base	64.2%	45.0%	12.6%	3.0%	50.0%	$778.24	126.0%
Bull	83.4%	51.7%	12.1%	3.5%	25.0%	$1,837.47	433.6%

Probability-weighted value **$893.42** (159.5% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 3,750.5 bn USD; terminal share 57.1%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **39.9%** vs consensus FY+1 64.2%.
- WACC (CAPM) = rf 5.2% + β 1.48 × ERP 5.0% = 12.6%.
- SOTP segments: AI semiconductors (XPU+networking) AI 半导体; Non-AI semis 非 AI 半导体; Infrastructure software (VMware) 基础软件 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

Customer in-sourcing (e.g., MediaTek on TPU), lumpy AI revenue, debt from VMware｜客户转单、AI 收入波动、VMware 债务 | Falsifiers: AI semi revenue guide decelerates; key XPU customer shifts vendor; VMware revenue declines｜AI 半导体指引放缓；主要 XPU 客户转供应商；VMware 收入下降

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `AVGO_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT - Version: v20261016-0756. Prior versions are never overwritten (see company history page).